

Live sandbox capstone

Browser labs taught the ideas, now you rehearse on a real GitHub template

Sandbox, not the curriculum repo

- The learn Git course tree is for reading modules and local examples
- The org template is your practice surface
- That gives you src, tb, scripts
- Read the sandbox note once so you know which URL is which

Capstone workflow

- Here is the end-to-end rhythm
- On GitHub, generate your repo from the `unix-git-practice` template
- Clone with `recurse submodules` so `shared-ip` is populated
- Change into the project and run `make test` to exercise the stub build and demo script
- Run the `check-ready` script before you push
- Create a lab branch, commit a small change, push to your fork

Real Git practice

```
wsl - module21-sandbox/examples/end_to_end

# Track A - real Unix
# real shell session (Track A)

# practice sandbox: /tmp/git-sbx-e4RNVv

$ git clone --recurse-submodules <template> my-practice

$ cd my-practice

/tmp/git-sbx-e4RNVv/my-practice

$ ls src tb scripts external/shared-ip
external/shared-ip:
README.md
VERSION
rtl

scripts:
README.md
check_ready.sh
run_demo.sh

src:
alu.v
main.v

tb:
test_main.v

$ make test
built build/main.o
INFO demo: starting
INFO demo: checking src/
INFO demo: checking tb/
WARN demo: no .env (copy .env.example)
INFO demo: submodule shared-ip present
INFO demo: PASS
make test: PASS

$ bash scripts/check_ready.sh
OK working tree clean
OK no build/logs tracked
OK .gitignore covers build/ and logs/
OK run_demo.sh exits 0
Ready to push.

$ git checkout -b lab1

$ git status
## lab1
```

Real shell — clone template, make test, lab branch

Real Git practice — try these

```
# git clone --recurse-submodules <your-template-url> my-practice  
git clone --recurse-submodules <your-template-url> my-practice
```

```
# cd my-practice — enter the cloned sandbox  
cd my-practice
```

```
# ls src tb scripts external/shared-ip — map the layout  
ls src tb scripts external/shared-ip
```

```
# make test — run stub build and demo checks  
make test
```

```
# bash scripts/check_ready.sh — pre-push readiness script  
bash scripts/check_ready.sh
```

```
# git checkout -b lab1 — branch for lab work
```

Pitfalls to watch

- Do not push assignments into the curriculum learn Git repository, use your template fork
- Always clone with recurse submodules or run submodule init afterward
- Run make and check-ready from the repo root, not a subdirectory
- And remember

Your turn

- Complete the sandbox checklist on your fork or an explore-only clone
- Clone with submodules, run make test
- When you are ready